# Andrew Bahsoun

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## Summary

Creative and disciplined undergraduate 4+1 EECS student motivated in pursuing research on trustworthy and fair machine learning systems for healthcare applications.

## Education

**Chapman University** *Aug 2022 – May 2027*
*BS Computer Science (4+1) + MS Electrical Engineering & Computer Science*

- **Selected coursework:** Data Structures & Algorithms; Programming Languages; Database Management; Software Design; Software Requirements & Testing; Computer Networks & Security; Scientific Computing; Deep Learning and Computer Vision

## Research & Professional Experience

**Research Assistant (Wen Lab)** *Orange, CA*
*Chapman University* *Aug 2024 – Present*

- Building a multimodal model to predict inpatient length-of-stay (LOS) using clinical time-series, notes, and imaging; implemented training and evaluation pipelines in Tensorflow.
- Exploring fairness-aware learning via constraint-based in-processing and adversarial approaches to reduce performance gaps across demographic groups.
- Working with large-scale EHR data, feature engineering, cohort filtering, and reproducible experiments on multi-GPU infrastructure.

**Software Engineer Intern** *Irvine, CA*
*Ingram Micro* *Jun 2025 – Aug 2025*

- Contributed within an Agile Software Development; practiced Test Driven Development with JUnit and improved reliability of backend services.
- Redesigned discount logic to support order-based discounts; updated OpenAPI schemas to enable a new service/engine data flow.
- Optimized production SQL queries and refactored legacy Java code by removing unnecessary logic and updating deprecated components.
- Built internal Python AI agents with Google ADK to audit and automate NodeJS dependency updates and answer internal questions using RAG retrieval; gathered requirements across technical and non-technical stakeholders.

**Summer Engineering Student Instructor** *Orange, CA*
*Chapman University* *June 2024 – July 2024*

- Coordinated weekly workshops to educate high schoolers on the fundamentals of machine learning, computer vision, and neural networks.
- Assembled Jupyter Notebook documents for classification models in TensorFlow. Presented lessons and assisted students through the ML building process.

**Tour Guide and Office Assistant** *Orange, CA*
*Chapman University* *June 2023 – July 2024*

- Lead campus tours for prospective students and families, presenting information about academic programs, campus life, and university resources.
- Support admissions operations and office coordination, contributing to prospective student engagement and recruitment efforts.

## Publications

- Inpatient Length of Stay and Mortality Prediction Utilizing Clinical Time Series Data — Junde Chen; Mason Li; Miles Milosevich; Tiffany Le; **Andrew Bahsoun**; Yuxin Wen. *IEEE Access*, 2025.
- Fairness-Constrained Multimodal Prediction of Hospital Length of Stay With Electronic Health Records. **Andrew Bahsoun**, Pinky Tran, Yuxin Wen. **Manuscript in preparation**, 2026.

## Funding & Awards

- **FIRE Grant** & **Robert A. Day Excellence Grant** (Chapman University). [Proposal]
- **Grand Prize / Overall Winner** — Chapman GCI Showcase (AI Handwashing Analyzer). [Article]

## Selected Projects

**AI Handwashing Analyzer** *[GitHub]*

- Built a hand-washing step recognition tool as a leader of a 6 person team using TensorFlow, and OpenCV.
- Deployed end-to-end workflow with data collection, model training, and evaluation; recognized as overall winner at Chapman University GCI Showcase.

**Canvas Notes Agent** *[GitHub] [Submission]*

- Built an agentic study assistant with a Chrome extension front-end that generates notes and study guides from Canvas course content.
- Implemented LangChain workflows and integrated Gemini-based LLM calls for summarization and Q&A; built for IEEE AI Dev Hack 2025.

**Lagrangian Constrained LOS Prediction** *(ongoing)*

- Multimodal LOS prediction with time-series vitals/labs, clinical notes embeddings, demographics, and X-ray features.
- Implemented with tensor fusion strategies and constrained loss on fairness metrics to limit demographic disparities.
- Built reproducible training runs with logging, hyperparameter sweeps, and GPU-aware execution on a shared cluster environment.

## Leadership & Service

**Camp Kesem Volunteer Counselor** *March 2023 – Present*
*Chapman University*

- Volunteered as a camp counselor creating a supportive and inclusive environment for children affected by a parent's cancer, emphasizing resilience, learning, and empathy.
- Mentored and communicated with children ages 5–15, helping them discuss their experiences and emotions related to their parent's illness.

**Chapman CARES Volunteer** *February 2023 – Present*
*Chapman University*

- Organized and participated in campus events such as the *Clothesline Project* and *Denim Day*, promoting dialogue on gender stereotypes and cultural beliefs.
- Helped train Chapman Resident Advisors on strategies to foster safe, healthy, and supportive residential communities.